5.5 Indefinite Integrals and the Substitution Method

The Fundamental Theorem of Calculus says that a definite integral of a continuous function can be computed directly if we can find an antiderivative of the function. In Section 4.8 we defined the **indefinite integral** of the function f with respect to x as the set of *all* antiderivatives of f, symbolized by $\int f(x)\,dx$. Since any two antiderivatives of f differ by a constant, the indefinite integral $\int$ notation means that for any antiderivative F of f,

$$\int f(x)\,dx = F(x) + C,$$

where C is any arbitrary constant. The connection between antiderivatives and the definite integral stated in the Fundamental Theorem now explains this notation:

$$\int_a^b f(x)\,dx = F(b) - F(a) = \left[F(b)+C\right] - \left[F(a)+C\right]$$
$$= \left[F(x)+C\right]_a^b = \left[\int f(x)\,dx\right]_a^b.$$

When finding the indefinite integral of a function f, remember that it always includes an arbitrary constant C.

We must distinguish carefully between definite and indefinite integrals. A definite integral $\int_a^b f(x)\,dx$ is a *number*. An indefinite integral $\int f(x)\,dx$ is a *function* plus an arbitrary constant C.

So far, we have only been able to find antiderivatives of functions that are clearly recognizable as derivatives. In this section we begin to develop more general techniques for finding antiderivatives of functions we can't easily recognize as derivatives.

① Some indefinite integral formulas

(a) $\int k\,dx = kx + C.$

(b) $\int x^n\,dx = \dfrac{1}{n+1}x^{n+1} + C.$

(c) $\int \sin kx\,dx = -\dfrac{1}{k}\cos x + C.$

(d) $\int \cos kx \, dx = \frac{1}{k}\sin x + C.$

② The differential

Let $y = f(x)$, then $dy = f'(x)dx$.

③ The Substitution method

We know that integration is the reverse process of differentiation. But all those functions

$$\sin(x^2)\,2x, \qquad (x^3 + x)^5\,(3x^2+1), \qquad x^3\cos(x^4 + 1)$$

don't have any indefinite formula for integration. So, the functions must be in simpler form with indefinite formula. For which, the substitution method is used. The substitution method in indefinite integral includes the substitution of a new variable in place of any small function in the given function.

- Substitution applies to integrals of form

$$\int f\left[g(x)\right] g'(x)\, dx$$

where $\int f(x)\,dx$ can be integrated by formulas in ①.

If we let $u = g(x)$, then

$$du = g'(x)dx.$$

Therefore, we get

$$\int f\left[g(x)\right] g'(x)\, dx = \int f(u)\, du.$$

Then, we integrate with respect to u and replace u by $g(x)$.

Example 1 Find $\int \sin(x^2)\,2x\,dx$.

Example 2 Find the integral $\int (x^3 + x)^5\,(3x^2+1)\,dx$.

Example 3 Find $\int \sqrt{2x+1}\ \mathrm{d}x$.

Example 4 Find $\int x^3 \cos(x^4+1)\,\mathrm{d}x$.

Example 5 Find $\int \cos(7x+3)\,\mathrm{d}x$.

Example 6 Evaluate $\int x\sqrt{2x+1}\ \mathrm{d}x$.

Example 7 Evaluate $\int \frac{2x\,\mathrm{d}x}{\sqrt[3]{x^2+1}}$.

④ The integrals of $\sin^2 x$ and $\cos^2 x$

Example 8 Evaluate $\int \sin^2 x\,\mathrm{d}x$ and $\int \cos^2 x\,\mathrm{d}x$.